Price Signal

Technical report - ModelX 2.0 Phase 2 implementation

Author: Nivetha Ramanan, Sri Lanka Institute of Information Technology **Domain:** Economy and livelihoods - cost of living and everyday economics **Repository:** https://github.com/IT25101414/price-signal **Release reviewed:** `v1.0.0-rc1`, plus the documented 18 October 2026 interface and validation updates **Data shown by the committed build:** Statistics Department, Central Bank of Sri Lanka Daily Price Report, 15 October 2026 **Public application:** `[PENDING - add the Render URL after owner-authorised deployment]`

1. Executive summary

Small canteens, restaurants and shops in Colombo can see today's wholesale and retail prices, but not whether an unusual move at Dambulla is likely to appear later at Pettah or Narahenpita. Price Signal converts the Central Bank of Sri Lanka's Daily Price Reports into a guarded estimate of how much of an unusual Dambulla-to-Colombo price gap may close in the next three report days.

The completed system has a polite, resumable PDF collector; a position-based table extractor; time-ordered models; calibrated uncertainty ranges; a fail-closed guard; daily refresh and publication scripts; a Streamlit dashboard; and a six-page Flask/SQLite website. The interface always states its data date and never says “buy” or “sell”. When the evidence is stale, unread, incomplete or too weak, the server removes the prices and estimate and gives a reason.

On held-out data from 2024 onward, adding Dambulla information improved forecasts beyond an own-history model in all three Colombo series. At three report days, the improvement in squared-error skill was 9.7 percentage points for wholesale Pettah, 8.3 for retail Pettah and 12.1 for retail Narahenpita. The stated 80% ranges contained 77.4% to 81.5% of unseen three-day outcomes. In a simulation, following the Dambulla model cost 1.22% to 1.57% less than always buying today. These are modest modelling results, not observed buyer savings.

A separate rolling-origin stress check refitted the model before three non-overlapping test windows from May to October 2026. Dambulla improved the own-history model in all nine market-window combinations by 5.5 to 13.7 points, while all nine wrong-item placebo gains were zero or negative. This supports recent temporal stability, but it is not evidence about the 2022 crisis.

The implementation is frozen and verified: 86 repository-owned automated tests pass with no environment-dependent skips; all 12 deliberate refusal cases pass; and the model, robustness and decision-policy synthetic self-tests pass. The 125-value check against original CBSL PDFs remains a separate explicit local validation because those source files are intentionally not redistributed. Public deployment remains pending external verification.

2. Proposal versus implementation

Proposal commitment	Result	Evidence
Respect robots.txt and the 10-second crawl delay	Complete	`download_cbsl_reports.py`, `data/manifest.csv`
Extract Daily Price Report tables and reject unknown layouts	Complete	`extract_prices.py`, `failed_files.txt`
Test whether Dambulla adds forecast value	Complete	`leadlag_check.py`, `robust_check.py`, saved result files
Compare with a placebo and own-history model	Complete	All saved lead-lag and robustness tables
Stress the result across later time windows	Complete	`rolling_stress_check.py`, `results_rolling_stress.txt`
Calibrate an 80% uncertainty range	Complete	`model.py`; 77.4%-81.5% coverage at three days
Test decision value	Complete as a simulation	`decision_backtest.py`; limitations stated
Refuse unsafe or unsupported output	Complete	`guard.py`; 12/12 refusal cases pass
Current user interface	Complete locally	Welcome, Today, Explore, Weekly report, How it works and Help
Daily cloud schedule	Deliberately replaced	`run_daily.sh`; the full generated price table is not republished
Gradient-boosting challenger	Completed, not adopted	Ridge retained under the pre-declared rule
Two or three buyer sessions	Complete	Three anonymous, consented sessions on 16-17 Oct; task results, ratings and limitations recorded
Public deployment	Prepared, pending owner sign-in	`render.yaml`, `web/DEPLOY.md`
Accessibility and public smoke checks	Implemented	Six-page accessibility check passes; local 8/8 smoke check passes; public workflow awaits the deployed URL

3. User, problem and delivered insight

The named user is a small Colombo food buyer who may decide whether to buy the full amount today, split a purchase or wait. CBSL reports show current prices and a short comparison, but they do not quantify whether the Dambulla market leads a selected Colombo market, how much an unusual gap tends to close or how uncertain that relationship is.

Price Signal delivers that missing comparison. For each supported food and Colombo series it shows the latest Dambulla and Colombo prices, the deviation from the usual market gap, the expected three-report-day Colombo change, an 80% likely range and a plain-language strength label. The project does not turn the estimate into purchasing advice. It gives evidence with its limits, leaving stock, spoilage, transport, cash flow and urgency to the buyer.

4. Data, access and temporality

The source is the Statistics Department, Central Bank of Sri Lanka, Daily Price Report. The collector discovered 2,274 manifest records through 15 October 2026; 2,273 have an `ok` download status and one is recorded as a failed PDF download. The supported analysis begins on 28 July 2020. Older layouts, scans without a text layer, undated headers and an exact duplicate are rejected rather than guessed.

The data are both historical and current. Years of history are required to learn and test the relationship, while the latest accepted report drives the user interface. `refresh.py` checks the newest listing, downloads and parses unseen reports, appends accepted data once and records each run. The committed cards and one-year public history slice end on 15 October 2026.

The source contains important irregularities: report layouts change; some cells are blank while others explicitly say “n.a.”; markets may be closed; item names drift; prices are rounded; and Monday reports often compare with Friday. The extractor keeps these states distinct and records the comparison basis. A disagreement between one report's summary and table is retained as a documented source issue.

CBSL replied in writing on 12 October 2026 that the published data are publicly available and do not require specific approval for use, asked for attribution to the Statistics Department and stated that it does not endorse or accept responsibility for derived findings. A redacted copy is committed in `docs/cbsl_reply_2026-10-12.pdf`. The raw PDF archive and complete generated price table are not redistributed. No personal data are used.

5. Architecture and stack

Data moves in one direction:

`CBSL listing and PDFs → refresh/collector → position-based parser → local price table → ridge model → safety guard → cards/SQLite → website and dashboard`

The guard is the only route to user-visible estimates. It runs when cards are built and again on every web request, because a previously safe card may become stale later.

Python 3.9 is used end to end. Requests and `urllib.robotparser` provide explicit crawling behaviour; pdfplumber exposes word positions required by the parser; pandas and NumPy support data preparation and the closed-form ridge models; Flask and SQLite serve a small inspectable web application; and pytest plus GitHub Actions provide repeatable checks. The public pages use plain HTML, CSS and JavaScript with no CDN or build step.

The web product contains:

- a plain-language Welcome page with a prominent independent-project notice;
- Today, with source health, summary counts, an uncertainty chart, filters, detailed cards and guarded CSV export;
- Explore, with paired Dambulla and Colombo history and their gap;
- Weekly report, comparing two complete five-report-day windows;
- How it works, presenting saved evaluation evidence and public-facing limitations; and
- Help, explaining labels, limitations and safe refusal.

6. Modelling method

The target is the log change in a Colombo series over one, three or five future report days. The production interface uses three. Model C uses the Colombo series' own one-day and five-day changes and its distance from its 20-day mean. Model D adds Dambulla's recent changes and the Dambulla-minus-Colombo gap relative to its trailing 60-day average. The difference in their performance measures what Dambulla adds beyond the price's own history.

Training and testing are ordered by time. Evaluation models fit data through 2023 and test on 2024 onward; labels that cross a split are dropped. Robustness intervals resample complete months, preserving local dependence. The live interval is split-conformal: residuals from a calibration period are scaled by recent item volatility, and their 80th percentile determines the likely range.

The “no clear signal” cutoff is determined on training data. A gradient-boosting challenger was evaluated under a rule fixed in advance: adopt it only when it improves three-day skill by at least 0.5 points in every Colombo series and its interval stays above zero. It failed that rule, so the simpler ridge model was retained.

7. Evaluation

7.1 Forecast value

The latest saved lead-lag run shows three-day squared-error skill gains of +9.7 points for wholesale Pettah, +8.3 for retail Pettah and +12.1 for retail Narahenpita. Gains are positive at one, three and five days. The robustness analysis' month-resampled intervals remain above zero while an item-swapped placebo is approximately zero. This supports a real but modest Dambulla lead.

The additional rolling-origin check uses only the committed recent-price slice and three chronological, non-overlapping test periods: 5 May-26 June, 30 June-21 August and 24 August-15 October 2026. The Dambulla-enhanced model beats the own-history control in all nine market-period combinations, with gains from +5.5 to +13.7 points. Every wrong-item placebo is zero or negative, and a majority of items benefit in all nine comparisons. Because this slice begins in October 2025, the result demonstrates recent stability rather than crisis-period robustness.

7.2 Honest uncertainty

At three report days, the nominal 80% ranges contain 81.0% of wholesale Pettah outcomes, 81.5% of retail Pettah outcomes and 77.4% of retail Narahenpita outcomes. All are within the proposal's 75%-85% acceptance band. Average half-widths are wide-18.1% to 28.1%-so the interface foregrounds the range rather than presenting the point estimate as certainty.

7.3 Simulated decision value

The simulated policy chooses whether to buy today or after three report days, using thresholds fixed before the test period. Against always buying today, the Dambulla model is 1.42% cheaper for wholesale Pettah, 1.22% cheaper for retail Pettah and 1.57% cheaper for retail Narahenpita. Directional accuracy across all calls is 62% to 69%, rising to 75% to 85% for the strongest quarter. The simulation assumes the purchase can be delayed without shortage, spoilage or operational cost, so it is not evidence of actual buyer savings.

7.4 Safe failure and software verification

Twelve constructed cases test fresh data, staleness, a newer unread report, Colombo or Dambulla closure, missing or `n.a.` prices, too little data, no clear signal and a holiday-adjusted freshness boundary. All 12 produce the specified state. Server tests verify that withheld cards and history contain no protected figures and that the interface never says “buy” or “sell”.

On 18 October 2026, default `pytest` reported 86 passed and zero skipped. Eleven repository-owned tests cover the extractor's cell parsing, row grouping, name cleaning and cross-report consistency without external fixtures; interface regression checks cover the explanations added after buyer feedback and the new rolling-evidence section. The stronger 125-value check against original source reports remains in `validate_extract_prices_real.py` and is invoked explicitly with local PDFs, rather than appearing as three environment-dependent skips in a fresh clone. The model, robustness and decision self-tests all pass on synthetic worlds with known answers. Python source compilation succeeds when its bytecode cache is directed to a writable project or temporary directory.

The release workflow now also runs the rolling stress check and a dependency-free accessibility regression. The accessibility check found no failures across the six pages and all tested light/dark colour pairs exceed the WCAG AA 4.5:1 contrast threshold. A rendered-browser structure audit found no unnamed visible controls, empty links, missing image alternatives, duplicate IDs, broken ARIA label references, positive tabindex values or heading-level jumps. A local service smoke test passed 8/8 checks across health, status and all public pages; the same test is prepared to run against the deployed URL on weekdays.

8. What the user sees and why it is trustworthy

Every page names the data source, and the live status makes the report date prominent. Direction is carried by signs, arrows and words rather than colour alone. The Today summary states how many estimates are available, which have clearer support, their likely direction and the average width of their possible ranges. Explore gives the latest Colombo-Dambulla difference, its typical level and the difference between them before showing the chart. The strongest-signal list is a way to navigate, not a claim that an estimate is certain.

If the source listing cannot be checked, if it lists a newer unread report, if the data exceed three working days, or if an item's inputs are incomplete, the website removes numbers server-side. Weekly comparisons require five observations in both weeks. Historical comparisons require dates shared by both series. These choices reduce apparent coverage, but they prevent a polished interface from hiding weak evidence.

Buyer evaluation and resulting changes

Three small-food buyers participated anonymously with consent: a restaurant buyer on 16 October and a vegetable-shop buyer and hotel/catering buyer on 17 October. Main-card understanding scores were 4, 3 and 4; trust as one buying input was 4, 2 and 4; usefulness was 5, 3 and 4. The median was 4/5 for all three measures. Two participants described using the site for checking or planning while still confirming the supplier's current price; the third would continue relying mainly on suppliers unless the signal's accuracy was clearer.

The task evidence exposed specific comprehension failures. Two participants needed help or could not explain the range; all three needed help or did not complete the historical comparison; and two questioned blank weekly values. The interface now describes the interval as a possible range that contained the real change about eight times in ten in past testing, explains shared dates and the start-at-100 comparison in plain language, states why weekly values are blank, and gives the source-data date its own status chip. An automated regression test confirms these explanations remain present. The revised wording has not yet been retested, and the exact tested commit, devices, setting and facilitator were not recorded, so this is formative evidence rather than proof of adoption or economic benefit.

9. Limitations and unresolved work

- The buyer study has only three participants and measures short-session comprehension and self-reported intent, not long-term use, purchasing behaviour or realised savings. The revised wording has not yet been retested.
- The public Render service has not yet been verified. The repository owner has begun the external service setup; the exact public URL will be recorded only after `/healthz` and the application pages respond successfully.

- The repository owner selected the MIT License on 18 October 2026. It covers the project's original software; CBSL data and reports, the Sri Lankan national emblem and other third-party material remain under their respective terms.
- The full generated table is local, so daily publication is manual. This avoids republishing the complete derived dataset until that choice is made explicitly.
- The study covers 18-19 foods with a Dambulla wholesale series, a calm 2024-2026 test period and one main train/test split.
- Report-day horizons span weekends and holidays; the current holiday file covers 2026 and needs extension.
- Prices are rounded, and large estimated moves can still have a range that crosses zero.

10. Reproduction

From a clean clone with Python 3.9 or later:

```
python3 -m pip install -r requirements.txt
python3 download_cbsl_reports.py
python3 extract_prices.py data/raw prices_all.csv
python3 robust_check.py prices_all.csv
python3 leadlag_check.py prices_all.csv
python3 decision_backtest.py prices_all.csv
python3 model.py prices_all.csv --evaluate
python3 model.py prices_all.csv
python3 web/build_db.py
python3 web/server.py
python3 -m pytest -q -rs
python3 refusal_report.py
```

The full download takes roughly six hours because the required ten-second delay is respected. To review the interface without rebuilding the full private working table, run `python3 web/server.py`; it uses the committed cards and one-year history slice.

11. AI assistance and responsibility

Claude (Anthropic) and OpenAI Codex assisted with research, first drafts of code and documents, testing, review, safety hardening, the redacted permission evidence and interface work. The file-by-file account is in `AI_USE.md`. The project author ran and reviewed the work and remains responsible for the submitted system and claims.

12. Submission completion rule

This report is final for the verified local release. After public deployment, replace the pending URL, regenerate the PDF from this source, record the video from that exact build and run the private-browser checklist.